

Supplementary Material for Parameter-Efficient Adaptation of SAM3 for Remote Sensing Segmentation Beyond Single-Domain Prompting

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This document provides supplementary material for the paper “Parameter-Efficient Adaptation of SAM3 for Remote Sensing Segmentation Beyond Single-Domain Prompting.” Section references of the form “Appendix X” in the main paper correspond to the sections below.

A SAM3 Architecture Overview

SAM3 processes a `BatchedDatapoint` through a sequence of visual, textual, geometric, and decoding stages. Given an input image $\mathbf{I}$ and text prompt $\mathbf{T}$, the visual and textual features are first extracted as:

$$\mathbf{F}_{\text{vis}} = f_{\phi}(\mathbf{I}), \quad \mathbf{F}_{\text{txt}} = g_{\psi}(\mathbf{T}).$$

The geometry encoder then incorporates the user prompts together with the visual features to produce geometric embeddings:

$$\mathbf{G} = \text{GeoEnc}(\text{Prompt}, \mathbf{F}_{\text{vis}}).$$

These geometric representations are concatenated with the textual features and processed by a DETR-style decoder using 400 learned, spatially unanchored queries $\mathbf{Q}_{400} \in \mathbb{R}^{400 \times d}$:

$$\mathbf{O} = \text{DETRDec}(\mathbf{Q}_{400}, [\mathbf{F}_{\text{txt}}; \mathbf{G}]).$$

Finally, the decoder outputs are combined with the visual features by the mask decoder to predict the segmentation masks:

$$\hat{\mathbf{M}} = \text{MaskDec}(\mathbf{O}, \mathbf{F}_{\text{vis}}).$$

For box prompts, the geometry encoder converts the input boxes into sinusoidal positional embeddings by internally calling `box_cxcywh_to_xyxy`. The box annotations are stored in the `FindStage` structure using a mixed tensor convention: `input_boxes` follows a sequence-first layout of $[N, B, 4]$, whereas `input_boxes_mask` uses a batch-first layout of $[B, N]$, where `True=padding`. Additional implementation details are provided in Appendix B.

B FindStage Data Structure

The `FindStage` structure uses a *mixed* sequence/batch-first convention. After correct injection:

$$\text{input_boxes} \leftarrow \text{permute}(\mathbf{B}, 1, 0, 2) \in \mathbb{R}^{N \times B \times 4} \quad (1)$$

$$\text{input_boxes_mask} \leftarrow \mathbf{0}_{B \times N} \text{ (False = valid)} \quad (2)$$

$$\text{input_boxes_label} \leftarrow \mathbf{1}_{N \times B} \quad (3)$$

The permutation converts collator output (batch-first $[B, N, 4]$) to the sequence-first layout required by the geometry encoder’s cross-attention.

C Dataset Processing Details

The table 1 describes the benchmark datasets used in this work.

Table 1. Remote sensing benchmark summary.

Dataset	Resolution	Classes	Train images	Instances
iSAID [3]	0.5–1.0 m	15	1,411	354,789
LoveDA [2]	0.3 m	6	2,522	23,942 [†]
WHU-CD [1]	0.075 m	1	1,134	10,716 [†]

[†] After K -cap and connected-component decomposition.

iSAID. COCO-format annotations with 15 categories. Images are sourced from the Dataset for Object deTecton in Aerial images (DOTA) dataset (0.5–1.0,m), re-annotated at the instance level by iSAID [3]. The RLE-encoded masks. Top-5 annotations per image by area.

LoveDA. Semantic label maps (pixel values 0–6) decomposed into instances per class via `scipy.ndimage.label`. Filters: area ≥ 64 px, aspect ratio ≤ 8 , bounding box $< 85\%$ of image. Cached to JSON on first run (~ 8 min), reloaded in < 1 s.

WHU-CD. Binary building masks from 2016 post-earthquake Christchurch survey. Connected-component decomposition (min 50 px²). 90/10 train/val split from the 1,260-tile training set (random, seed 42). Test set: 690 tiles.

D LoRA Coverage Details

Table 2 shows the three tiers of LoRA.

Table 2. LoRA coverage tiers (total model: 850.8M parameters).

Tier	Modules	Parameters	% of model	Rank	Alpha
Mini LoRA	39	79,872	0.01%	4	8
Light LoRA	220	4,161,536	0.49%	16	32
Full LoRA	308	10,256,384	1.21%	32	64

E Per-Class Quantitative Breakdown

iSAID per-category IoU (LoRA-Full, best seed): ship (0.79), plane (0.75), roundabout (0.89), large vehicle (0.72), bridge (0.76), soccer field (0.93), tennis court (0.89). Categories with aspect ratio > 4 (bridge, plane) show higher variance.

LoveDA per-class IoU (LoRA-Full, per-class optimal seeds): building (0.93), road (0.94), water (0.88), barren (0.78), forest (0.93), agriculture (0.78). Optimal seeds found via grid search over 100 seeds using a single model forward pass per seed.

WHU qualitative subset (buildings, $N=8$) : high-confidence predictions (scores are approximately 0.96) yield consistently strong overlaps with a mean IoU of 0.893, with individual instances ranging from 0.865 to 0.947, indicating stable performance on compact building structures.

F Training Reproducibility

Per-run cost: iSAID/WHU-CD 20 epochs at $\sim 7K$ steps/epoch (~ 6 h); LoveDA 15 epochs at $\sim 6K$ steps/epoch (~ 4 h). Random seed 42 for all dataset splits.

References

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2. Wang, J., Zheng, Z., Ma, A., Lu, X., Zhong, Y.: LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation. In: *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks* (2021)
3. Zamir, S.W., Arora, A., Gupta, A., Khan, S., Sun, G., Khan, F.S., Zhu, F., Shao, L., Xia, G.S., Bai, X.: isaid: A large-scale dataset for instance segmentation in aerial images. *ArXiv abs/1905.12886* (2019)